11.1 Four different teaching methods are compared. Test scores for students in each methods:

method I: 82, 85, 79, 88, 84

method II: 75, 78, 72, 80, 74.

method III: 88, 92, 86, 90, 89

method IV: 70, 73, 68, 72, 71

a. Perform a Kruskal-Wallis test at $\alpha = 0.05$.

b. If significant, conduct post-hoc pairwise comparison using Dunn's test with Bonferroni correction.

H_0: All 4 samples are drawn from populations having the same location (median)

H_1: The samples are drawn from populations with different medians.

score.	68	70	71	72	72	73	74	75	78	79	80	82	84	85
method.	IV	IV	IV	IV	II	IV	II	II	II	I	II	I	I	I
rank.	1	2	3	4.5	4.5	6	7	8	9	10	11	12	13	14

score	86	88	88	89	90	92
method	III	III	I	III	III	III
rank.	15	16.5	16.5	18	19	20

$R_I = 65.5$ $\quad N = 20$.

$R_{II} = 39.5$. $\quad n_1 = n_2 = n_3 = n_4 = 5$.

$R_{III} = 88.5$ $\quad k = 4$.

$R_{IV} = 16.5$.

The Kruskal Wallis Test Statistic, $H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(N+1)$

$$H = \frac{12}{20(20+1)} \left[\frac{65.5^2 + 39.5^2 + 88.5^2 + 16.5^2}{5} \right] - 3(20+1)$$

$H = 16.74$ (4 sf)

Adjustment of ties:

$$H_c = \frac{H}{1 - \frac{\sum_{j=1}^{g}(t_j^3 - t_j)}{N^3 - N}}$$

where g is the no. of tied groups.
t_j is the size of the j-th tied group.

$\Rightarrow g = 2.$
$t_j = 2.$

$$\sum_{j=1}^{2}(t_j^3 - t_j) = (2^3-2) + (2^3-2) = 12.$$

$$H_c = \frac{16.74}{1 - \frac{12}{20^3 - 20}} = 16.77 \text{ (4 sf)}.$$

Under H_0, H follows an approximate chi-square distribution with $k-1$ degrees of freedom.

$$H_{critical} = 7.815.$$

Since $H_c > H_{critical}$, we reject the null hypothesis and conclude a significant difference in the location of the four sampled populations. We therefore proceed to conduct the Dunn's test with a Bonferroni correction for pairwise comparison.

The Dunn's Test:

$$Z = \frac{|\bar{R}_i - \bar{R}_j|}{\sqrt{\frac{N(N+1)}{12}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}}$$

For any two samples i & j; $\bar{R}_{i,j} = \frac{R_{i,j}}{n_{i,j}}$

$\bar{R}_I = 13.1$ $\bar{R}_{II} = 7.9$ $\bar{R}_{III} = 17.7$ $\bar{R}_{IV} = 3.3.$ $\sqrt{\frac{20(20+1)}{12}\left(\frac{1}{5} + \frac{1}{5}\right)} = \sqrt{14}.$

$$Z_{I,II} = \frac{|13.1 - 7.9|}{\sqrt{14}} = 1.3898 \text{ (4 sf)}. \qquad Z_{I,III} = \frac{|13.1 - 17.7|}{\sqrt{14}} = 1.229 \text{ (4 sf)}.$$

$Z_{I,IV} = \dfrac{|13.1 - 3.3|}{\sqrt{14}} = 2.619$ (4sf). $Z_{II,III} = \dfrac{|7.9 - 17.7|}{\sqrt{14}} = 2.619$ (4sf)

$Z_{II,IV} = \dfrac{|7.9 - 3.3|}{\sqrt{14}} = 1.229$ (4sf). $Z_{III,IV} = \dfrac{|17.7 - 3.3|}{\sqrt{14}} = 3.849$ (4sf)

The Bonferroni correction adjusts the significance level using the following formula.

$$\alpha_{adj} = \frac{\alpha}{{}_kC_2} \qquad \text{now } \alpha = 0.05,\ K = 4.$$

$$\alpha_{adj} = \frac{0.05}{\binom{4}{2}} = 0.0083$$

The adjusted p-value of 0.0083 is obtained from an absolute z-value of 2.415.

$\Rightarrow$ $2.415 > Z_{I,IV}, Z_{II,III}, Z_{III,IV}$

Therefore the pairwise comparison of method I and IV, II and III and III and IV are significant.

In conclusion, the Kruskal Wallis test informs us of a statistical significant difference in location of the four methods of teaching. A posthoc Dunn's test with a Bonferroni correction further informs us of the significant pairwise location difference present between the methods; I and IV

II and III

III and IV.

11.2. Six participants rate ē ir satisfaction (1-10 scale) with 3 different smartphone interfaces.

participant	A	B	C
1	8	6	7
2	7	5	8
3	9	7	6
4	6	8	9
5	8	5	7
6	7	6	8.

a. Perform a Friedmann test at $\alpha = 0.05$.

b. If significant, ~~so~~ conduct a post-hoc analysis using WSR tests with Bonferroni correction.

The friedman test

- This is a nonparametric test, ~~altern~~ an alternative to repeated measures ANOVA, extending ē sign test to multiple related samples.

Assumptions
1. The data consists of n subjects (blocks) and k treatments (conditions).
2. Observations are measured on at least an ordinal scale.
3. The ~~blocks~~ subjects are independent of each other.
4. Within each subject (block) observations can be rank.

Hypotheses: H_0: All k treatments have ē same effect (identical distribution)

vs.

H_1: At least one treatment tends to yield larger values ~~that~~ than at least one other.

Example:

participant	A	rank	B	rank	C.	rank.
1	8	3	6	1	7.	2
2	7	2	5	1	8	3
3	9	3	7	2	6	1
4	6	1	8	2	9	3
5	8	3	5	1	7	2
6.	7	2	6	1	8	3

$\sum R_A = 14.$ $\sum R_B = 8$ $\sum R_C = 14.$

n(subjects) = 6.
k(treatments) = 3.
$R_A = 14$
$R_B = 8$
$R_C = 14.$

$$F_r = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1).$$

NB: Under H_0, F_r follows approximately a chi-square distribution with $(k-1)$ degrees of freedom for moderate to large n.

Adjustment for ties:
When they occur, within subjects, assign avg. ranks & apply a correction.

$$F_r = \frac{12\sum_{j=1}^{k} R_j^2 - 3n^2k(k+1)^2.}{nk(k+1) + \frac{1}{k-1}\sum_{i=1}^{n}\sum_{j=1}^{k}(r_{ij}^3 - r_{ij})}$$

where r_{ij} are ē ranks within subject i.

$$F_r = \frac{12}{18 \times 4}\left[14^2 + 8^2 + 14^2\right] - 3 \times 6\left[4\right].$$

$F_{r\,calc} = 4.$

Under H_0, F_r follows an approximate chi-square distribution with $k-1$ degrees of freedom for moderate to large sizes of n.

$$\chi^2_{3-1, 0.05} = 5.99 \ \{\text{critical value for rejection of } H_0\}$$

Since $F_r < \chi^2_{2, 0.05}$, we fail to reject the null hypothesis. In conclusion all 3 smartphone interfaces offer the same satisfaction level. Post-hoc tests are therefore not needed in this case.

11.3. A study examines the effects of a drug dose on pain relief. Four dose levels (0mg, 5mg, 10mg, 20mg) are tested, with 6 subjects per group. Pain relief scores (higher = better)

0mg	5mg	10mg	20mg
2	4	5	7
3	3	6	6
1	5	4	8
2	4	5	7
3	3	6	6
2	4	5	7

a. Test for an increasing trend using the Jonckheere-Terpstra test.

b. Compare your results with a Kruskal-Wallis test.

c. Which test is more appropriate and why?

0mg	Rank	5mg	Rank	10mg	Rank.	20mg	Rank.
2	3	4	10.5	5	14.5	7	22
3	6.5	3	6.5	6	18.5	6	18.5
1	1	5	14.5	4	10.5	8	24
2	3	4	10.5	5	14.5	7	22
3	6.5	3	6.5.	6	18.5.	6	18.5
2	3	4	10.5	5	14.5	7	22

H_0: The 4 dose distributions are identical

vs.

H_1: The distributions are ordered in a specified direction.

$$JT = \sum_{i=1}^{k-1} \sum_{j=1}^{k} U_{ij}$$

$$E(JT) = \frac{N^2 - \sum_{i=1}^{k} n_i^2}{4}$$

$$Var(JT) = \frac{N^2(2N+3) - \sum_{i=1}^{k} n_i^2(2n_i+3)}{72}$$

0mg - 5mg, 0mg - 10mg, 0mg - 20mg. 0mg < 5mg <

5mg - 10mg, 5mg - 20mg.

10mg - 20mg

0mg vs 5mg

0mg \ 5mg	4	3	5	4	3	4
2	1	1	1	1	1	1
3	1	0	1	1	0	1
1	1	1	1	1	1	1
2	1	1	1	1	1	1
3	1	0	1	1	0	1
2	1	1	1	1	1	1

32.

0mg vs 10mg

0mg. \ 10mg.	5	6	4	5	6	5
2	1	1	1	1	1	
3	1	1	1	1	1	
1	1	1	1	1	1	
2	1	1	1	1	1	
3	1	1	1	1	1	
2	1	1	1	1	1	

36.

0mg - 20mg.

36.

5mg - 10mg.

5mg. \ 10mg	5	6	4	5	6	5
4	1	1	0	1	1	1
3	1	1	1	1	1	1
5	0	1	0	0	1	0
4	1	1	0	1	1	1
3	1	1	1	1	1	1
4	1	1	0	1	1	1

30.

5mg - 20mg

36.

10mg - 20mg.

10mg. \ 20mg.	7	6	8	7	6	7
5	1	1	1	1	1	1
6	1	0	1	1	0	1
4	1	1	1	1	1	1
5	1	1	1	1	0	1
6	1	0	1	1	1	1
5	1	1	1	1	1	1

= 32.

$JT = 32 + 36 + 36 + 30 + 36 + 32 = 202.$

$N = 24.\ k = 4.$

$\forall_i : i = 1, 2, 3, 4 \quad n_i = 6.$

$$E(JT) = \frac{24^2 - 4(6^2)}{4} = 108$$

$$Var(JT) = \frac{24^2[2(24)+3] - 2160}{72} = 378$$

$$Z = \frac{JT - E(JT)}{\sqrt{Var(JT)}} = \frac{202 - 108}{\sqrt{378}} = 4.835.$$

$\Rightarrow p < 0.00003$

~~we reject the null hypothesis in favour of the alt~~

For the Jonckheere-Terpstra test, we reject the null hypothesis in favour of the claim that the four doses are ordered in a specified direction. (trend)

For the Kruskal-Wallis Test, H_0: location of the 4 doses is the same.
H_1: location of the four doses ~~is~~ differs.

$R_{0mg} = 23$ $R_{5mg} = 59$ $R_{10mg} = 91$ $R_{20mg} = 127$

$$H = \frac{12}{24(25)} \cdot \left[\frac{23^2 + 59^2 + 91^2 + 127^2}{6}\right] - 3(24+1) = 19.73 \text{ (4sf)}.$$

Adjusting ties:

$g = 6.$ $\quad \sum_{j=1}^{6} (t_j^3 - t_j) = (3^3 - 3) + 4(4^3 - 4) + (3^3 - 3)$
$= 288$

$$H_c = \frac{19.73}{1 - \frac{288}{24^3 - 24}} = 20.15 \text{ (4sf)}$$

$$\chi^2_{3,0.05} = 7.81$$

Since $H_c > \chi^2_{3,0.05}$, we reject the null hypothesis in favour of the claim that the median ~~value of the~~ pain-relief time of the four doses is different.

The Jonckheere-Terpstra test is a better alternative to the Kruskal Wallis test when dealing with ~~observations~~ samples obtained from ordered ~~samples~~ populations.

11.4 Using the data from Exercise 1, perform a median test to compare the four teaching methods. Compare the results with a Kruskal-Wallis test. Which test do you think is more valid?

I	II	III	IV
82	75	88	70
85	78	92	73
79	72	86	68
88	80	90	72
84	74	89	71

H_0: Population medians for all methods are equal.

H_1: Not all population medians for all methods are equal.

The overall median = 79.5

> Since k=4, we generate a 2×4 contingency table for values (scores) ~~the~~ below & above the median as below.

~~above~~.

	I	II	III	IV	
above	1	4	0	5	10
below	4	1	5	0	10
	5	5	5	5	20

We calculate the chi-square statistic for the median test involving multiple samples.

$$\chi^2 = \sum_{i=1}^{2}\sum_{j=1}^{4} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \quad \text{where } E_{ij} = \frac{\text{row total} \times \text{column total}}{N}$$

$$\forall j \quad E_{1j} = \frac{5 \times 10}{20} = 2.5.$$

$$\forall i \quad E_{2j} = \frac{5 \times 10}{20} = 2.5.$$

$$\chi^2 = \frac{(1-2.5)^2}{2.5} + \frac{(4-2.5)^2}{2.5} + \frac{(0-2.5)^2}{2.5} + \frac{(5-2.5)^2}{2.5} +$$

$$\frac{(4-2.5)^2}{2.5} + \frac{(1-2.5)^2}{2.5} + \frac{(5-2.5)^2}{2.5} + \frac{(0-2.5)^2}{2.5} = 8.6.$$

$\chi^2_{calc} = 8.6.$

$\chi^2_{4-1,0.05} = 7.81 = \chi^2_{critical}.$

Since $\chi^2_{calc} > \chi^2_{critical}$, we reject the null hypothesis.

The Kruskal Wallis test in Exercise 1 reported a 5% significance level difference in the four teaching methods. The Kruskal Wallis test is much more valid in this case as there are no outlier observations present.

11.5. Eight patients try three different treatments for insomnia and report whether they slept better (1) or not (0).

patient	tr. A	tr. B	tr. C.	L_i
1	1	0	1	2
2	0	1	1	2
3	1	0	0	1
4	0	0	1	1
5	1	1	1	3
6	0	0	0	0
7	1	0	1	2
8	0	1	1	2
G_j	4	3	6	13.

a. Perform a Cochran's Q test at $\alpha = 0.05$.

b. If significant, conduct pairwise comparisons using McNemar's test with Bonferroni's correction.

c. Interpret your findings.

Test Procedure.

- Let X_{ij} be the response (0 or 1) for block i, treatment j. Let

$$G_j = \sum_{i=1}^{n} X_{ij}$$ be the column total for treatment j

$$L_i = \sum_{j=1}^{k} X_{ij}$$ be the row total for block i. 11.5

$$N = \sum_{j=1}^{k} G_j = \sum_{i=1}^{n} L_i$$ be the grand total.

The test statistic is defined as;

$$Q = \frac{k(k-1)\sum_{j=1}^{k}\left(G_j - \frac{N}{k}\right)^2}{\sum_{i=1}^{n} L_i(k-L_i)}$$

- Under H_0 (treatments equally effective), Q follows approximately a chi-square distribution with k-1 degrees of freedom.

$k=3$. $n=8$

$N=13$.

$$Q = \frac{3(2)\left[\left(4-\frac{13}{3}\right)^2+\left(3-\frac{13}{3}\right)^2+\left(6-\frac{13}{3}\right)^2\right]}{\left[2(3-2)+2(3-2)+1(3-1)+1(3-1)+\cancel{3(3-3)}^{0}+\cancel{0(3-0)}^{0}+2(3-2)+2(3-2)\right]}$$

$$Q = \frac{\frac{14}{3}\times 6}{12} = \frac{28}{12} = 2.333\ (4sf).$$

$$\chi^2_{2,0.05} = 5.99.$$

Since $2.333 < \chi^2_{2,0.05}$, we fail to reject H_0. There's no evidence that medication differs.

MAS 812:Non-Parametric Methods

continued

Gift Pamba

MSC/MAT/00098/024

EXERCISE 11.6

Generate data for three groups with distributions:

- Group 1: Normal(0,1)
- Group 2: Normal(0.5, 1)
- Group 3: Normal(1,1)

with *n=10* per group.

A. Perform a one-way ANOVA, Kruskal-Wallis and median test

i. ANOVA

```
## One-way ANOVA
## p-value: 0.3242
## Decision: Fail to reject the null hypothesis
```

ii. Kruskal-Wallis

```
## Kruskal-Wallis test
## p-value: 0.2241
## Decision: Fail to reject the null hypothesis
```

iii. Median test

```
## Median test
## p-value: 0.213
## Decision: Fail to reject the null hypothesis
```

iv. Conclusion

For all the three tests, we fail to reject the null hypotheses. There is sufficient evidence in favor of the initial claim of an insignificant difference between the three groups at 5% level.

B. Repeat 1000 times and compute empirical power for each test

```
## Empirical power - Normal values
## One-way ANOVA: 0.463
## Kruskal Wallis test: 0.437
## Median Test: 0.253
```

C. Repeat with Cauchy distributions (heavy-tailed) and compare

For this simulation, I use the following Cauchy distributions with 10 observations each;

- Cauchy(location = 0, scale = 1)
- Cauchy(location = 0.5, scale = 1)
- Cauchy(location = 1, scale = 1)

```
## Empirical power - Cauchy values
## One-way ANOVA: 0.049
## Kruskal Wallis test: 0.172
## Median Test: 0.162
```

	Normal	Cauchy
One-way ANOVA	0.463	0.049
Kruskal-Wallis test	0.437	0.172
Median test	0.253	0.162

D. What do you conclude about the relative power of these tests?

The One-way ANOVA test is robust for normally distribted observations with equal variances. However, it's power is greatly affected by outliers as shown by it's power when it tests heavily tailed distributions.

The Kruskal Wallis test is less powerful than the One-way ANOVA test when dealing with symmetrically distributed data. However, it is a better alternative if the observations are riddled with outliers.

The median test possesses the least power in comparison to the two tests. This is due to the information loss attributed to the convertion of continuous observations to qualitative frequencies based on the median value.

EXERCISE 11.7

A. Why is it important to follow a significant Kruskal-Wallis test with post-hoc comparisons rather than simply reporting the omnibus result?

Significant Kruskal-Wallis test reporting is vague in nature. Post-hoc comparisons solve this by pinpointing the pairwise groups that differ significantly.

B. Explain the difference between the Kruskal-Wallis test and the Friedman test in terms of design and assumptions

In terms of design, observations from different samples under the Kruskal-Wallis test are independent of each other whereas under the Friedman test observations are of repeated measures. Under the Kruskal-Wallis test ranking is conducted across all the samples while for the Friedman test, ranking is done within each block/participant.

C. Under what circumstances would you choose the Jonckheere-Terpstra test over the Kruskal-Wallis test?

The Jonckheere-Terpstra test would be appropriate if the researcher is interested in discovering an ordered difference amongst the groups under investigation.

D. What is the purpose of Bonferroni correction in post-hoc analysis, and what is its limitation?

Bonferroni correction identifies significant pairwise differences present between quantitative values obtained from different or similar populations.

Limitation: The correction may overlook significant pairwise differences due to a reduced margin of error attributed to an increase in the number of samples under comparison.

E. Why is the median test generally less powerful than the Kruskal-Wallis test for detecting location differences?

Under the median test, quantitative observations are converted to frequency values less or above the overall median value, while the Kruskal-Wallis test makes use of the observations' ranks. This information loss makes the median test a less powerful alternative to the Kruskal-Wallis test for detecting location differences.